

Latihan TM06 MBD

Nama: Gilbran Mahdavia Raja

NIM: 5025241134

No.1

Sintaks:

```
CREATE VIEW view_customer_salary_above_2000 AS

SELECT name, salary, age

FROM customers

WHERE

    salary > 2000;
```

Hasil:

		name	salary	age
				integer
☐	>	Chaitali	6500.00	25
☐	>	Hardik	8500.00	27
☐	>	Komal	4500.00	22

No.2

Sintaks:

```
CREATE VIEW view_salary_percentage AS

SELECT name, (

    salary / (
```

```

        SELECT SUM(salary)
        FROM customers
    )
    ) * 100 AS percentage
FROM customers
ORDER BY percentage DESC;

```

Hasil:

☐	Q	name	percentage
☐	>	Hardik	34.000000000000000000
☐	>	Chaitali	26.000000000000000000
☐	>	Komal	18.000000000000000000
☐	>	Ramesh	8.000000000000000000
☐	>	Kaushik	8.000000000000000000
☐	>	Khilan	6.000000000000000000

No.3

Sintaks:

```

CREATE TABLE salary_log (
    IDLog SERIAL PRIMARY KEY,

```

```

        log_date TIMESTAMP,
        previous_salary NUMERIC(10, 2),
        next_salary NUMERIC(10, 2)
    );

CREATE OR REPLACE FUNCTION log_salary_change()
RETURNS TRIGGER AS $$
BEGIN
    IF OLD.salary <> NEW.salary THEN
        INSERT INTO salary_log (log_date, previous_salary,
next_salary)
            VALUES (NOW(), OLD.salary, NEW.salary);
    END IF;
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_salary_update
AFTER UPDATE ON customers
FOR EACH ROW
EXECUTE FUNCTION log_salary_change();

```

Hasil:

	idlog integer	log_date	previous_salary	next_salary
> 1		2026-04-07 17:11:11.30	2000.00	3000.00

No.4

Sintaks:

```
CREATE OR REPLACE FUNCTION enforce_salary_range()
RETURNS TRIGGER AS $$
BEGIN
    IF NEW.salary > 8500 THEN
        NEW.salary := 8500;
    ELSIF NEW.salary < 1500 THEN
        NEW.salary := 1500;
    END IF;
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

-- trigger untuk INSERT

CREATE TRIGGER trg_salary_range_insert
BEFORE INSERT ON customers
FOR EACH ROW
EXECUTE FUNCTION enforce_salary_range();
```

```
-- trigger untuk UPDATE

CREATE TRIGGER trg_salary_range_update
BEFORE UPDATE ON customers
FOR EACH ROW
EXECUTE FUNCTION enforce_salary_range();
```

Hasil:

```
INSERT INTO customers VALUES ( 7, 'Test', 30, 'Surabaya', 9000 );
-- jadi 8500

INSERT INTO customers VALUES ( 8, 'Test2', 28, 'Malang', 1000 );
-- jadi 1500
```

		id integer	name	age integer	address	salary
	>	2	Khilan	25	Delhi	1500.00
	>	3	Kaushik	23	Kota	2000.00
	>	4	Chaitali	25	Mumbai	6500.00
	>	5	Hardik	27	Bhopal	8500.00
	>	6	Komal	22	MP	4500.00
	>	1	Ramesh	32	Ahmedabad	3000.00
	>	7	Test	30	Surabaya	8500.00
	>	8	Test2	28	Malang	1500.00